Activity-APPLICATION BUILDING USING WEKA TOOL

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1. Weather-Based Play Prediction

Build a WEKA application that predicts whether a person should play outdoors based on weather conditions.

- Dataset: Outlook, Temperature, Humidity, Windy
- Target: Play (Yes/No) • Algorithm: **J48 Decision Tree**
- Tasks:
  1. Load the dataset into WEKA.
  2. Preprocess the data.
  3. Apply J48.
  4. Visualize the decision tree.
  5. Predict Play for a new weather condition.

2. Flower Species Classification

Build a WEKA application that identifies the species of a flower from its physical measurements.

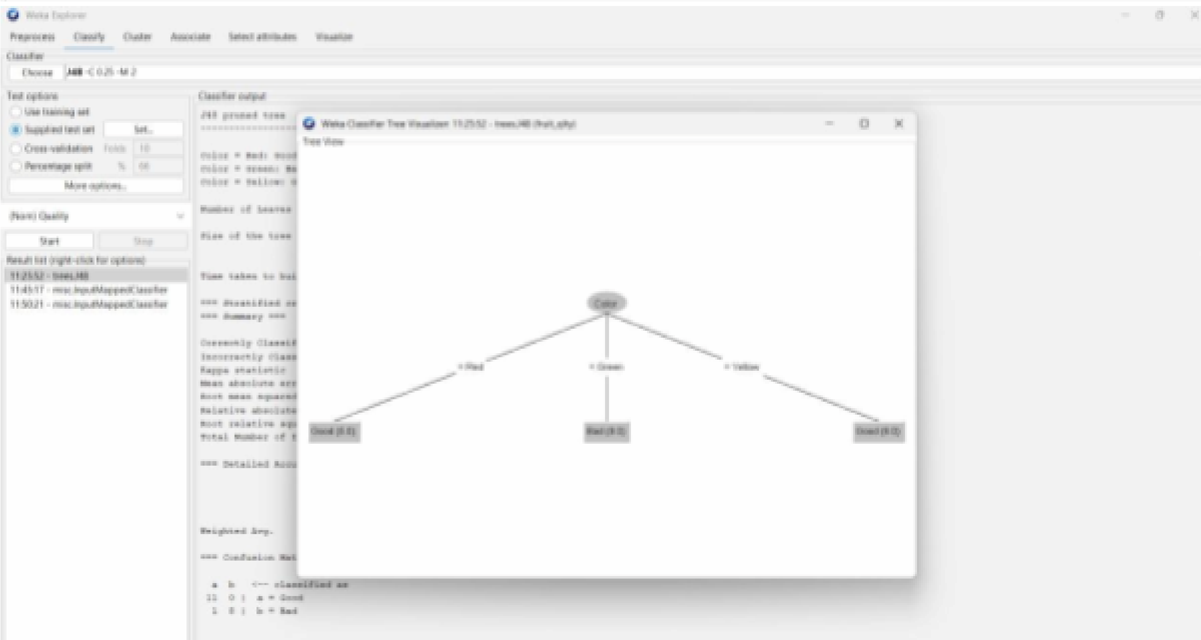
- Dataset: Iris dataset
- Attributes: Sepal Length, Sepal Width, Petal Length, Petal Width
- Target: Iris Species
- Algorithm: **J48 or Naive Bayes**
- Tasks:
  1. Load the Iris dataset.
  2. Select the classification algorithm.
  3. Train the model.
  4. Evaluate its accuracy.
  5. Classify a new flower.

Answers:

1) **Weather-Based Play Prediction**

Classifier output

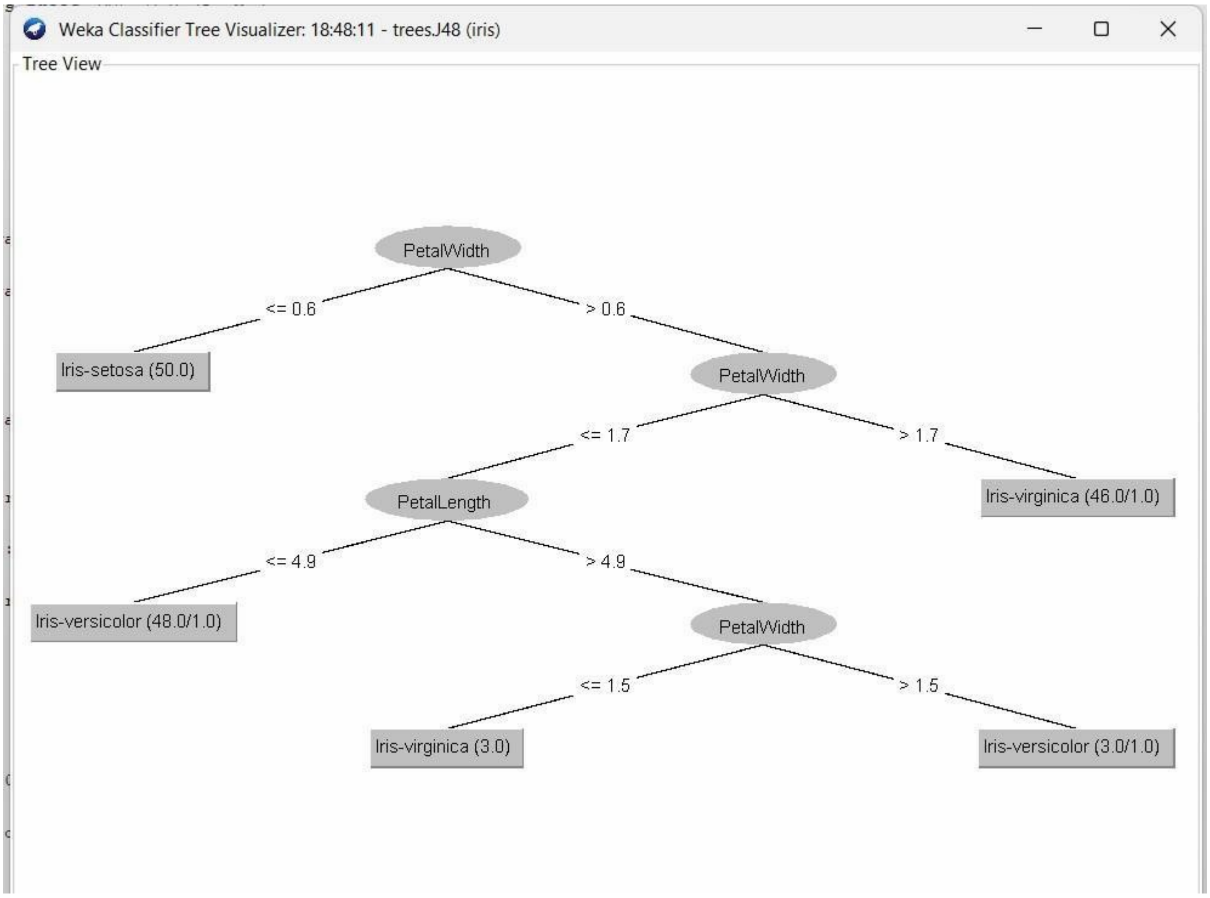
```
Model attributes            Incoming attributes
--------------------        ----------------
(numeric) Weight        --> 1 (numeric) Weight
(nominal) Color         --> 2 (nominal) Color
(nominal) Size          --> 3 (nominal) Size
(numeric) Firmness      --> 4 (numeric) Firmness
(nominal) Quality       --> 5 (nominal) Quality


Time taken to build model: 0 seconds

=== Predictions on test set ===

inst#,actual,predicted,error,prediction
1,1:?,1:Good,,1
2,1:?,2:Bad,,1

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0.01 seconds

=== Summary ===

Total Number of Instances                0
Ignored Class Unknown Instances                  2

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 ?        ?        ?          ?        ?          ?        ?         ?         Good
                 ?        ?        ?          ?        ?          ?        ?         ?         Bad
Weighted Avg.    ?        ?        ?          ?        ?          ?        ?         ?

=== Confusion Matrix ===

 a b   <-- classified as
 0 0 | a = Good
 0 0 | b = Bad
```



2) **Flower Species Classification**

Classifier output

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:     iris
Instances:    150
Attributes:   5
              SepalLength
              SepalWidth
              PetalLength
              PetalWidth
              Class
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
------------------

PetalWidth <= 0.6: Iris-setosa (50.0)
PetalWidth > 0.6
|   PetalWidth <= 1.7
|   |   PetalLength <= 4.9: Iris-versicolor (48.0/1.0)
|   |   PetalLength > 4.9
|   |   |   PetalWidth <= 1.5: Iris-virginica (3.0)
|   |   |   PetalWidth > 1.5: Iris-versicolor (3.0/1.0)
|   PetalWidth > 1.7: Iris-virginica (46.0/1.0)

Number of Leaves  :     5

Size of the tree :      9


Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         144               96      %
Incorrectly Classified Instances         6                4      %
Kappa statistic                          0.94
Mean absolute error                      0.035
Root mean squared error                  0.1586
Relative absolute error                  7.8705 %
Root relative squared error             33.6353 %
Total Number of Instances              150

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.980    0.000    1.000      0.980    0.990      0.985    0.990     0.987     Iris-setosa
                 0.940    0.030    0.940      0.940    0.940      0.910    0.952     0.880     Iris-versicolor
                 0.960    0.030    0.941      0.960    0.950      0.925    0.961     0.905     Iris-virginica
Weighted Avg.    0.960    0.020    0.960      0.960    0.960      0.940    0.968     0.924

=== Confusion Matrix ===

  a  b  c   <-- classified as
 49  1  0 |  a = Iris-setosa
  0 47  3 |  b = Iris-versicolor
  0  2 48 |  c = Iris-virginica
```


Weka Classifier Tree Visualizer: 18:48:11 - trees.J48 (iris)
Tree View
PetalWidth
<= 0.6
> 0.6
Iris-setosa (50.0)
PetalWidth
<= 1.7
> 1.7
PetalLength
Iris-virginica (46.0/1.0)
<= 4.9
> 4.9
Iris-versicolor (48.0/1.0)
PetalWidth
<= 1.5
> 1.5
Iris-virginica (3.0)
Iris-versicolor (3.0/1.0)